

TAK, Thailand

WMO: 483760

Lat: 16.8797N

Lon: 99.1400E

Elev: 415

StdP: 14.48

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	58.5	61.4	48.2	50.7	83.8	51.1	56.6	81.5	9.1	83.1	7.7	82.7	0.1	270	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	19.6	103.4	75.6	101.3	75.1	99.3	74.8	81.1	91.1	80.5	90.4	79.9	89.7	4.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.9	152.6	85.2	78.2	148.5	84.5	77.4	144.9	83.8	45.3	91.7	44.5	90.4	43.9	90.1	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.9	9.4	53.8	106.7	3.0	1.9	51.6	108.1	49.8	109.2	48.1	110.3	45.9	111.7		
(5)			52.1	83.3	3.0	2.1	49.9	84.8	48.2	86.1	46.5	87.2	44.3	88.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.0	76.8	81.7	86.8	90.0	86.9	84.9	83.7	83.5	83.5	82.0	79.6	76.3		
	DBStd	5.23	4.06	3.96	4.50	4.13	4.42	2.77	2.50	2.29	2.29	2.38	3.24	4.07		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	1	1	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	12036	831	886	1141	1201	1145	1047	1045	1038	1005	992	888	817		
	CDD65	6562	367	466	676	751	680	597	580	573	555	527	438	352		
	CDH74	76344	3589	5617	9278	10777	8717	7050	6539	6352	6055	5277	4031	3062		
(13)	CDH80	36353	1602	3048	5471	6710	4610	3004	2459	2302	2327	1998	1622	1200		
(14)	Wind	WSAvg	2.2	1.0	1.6	2.4	3.2	3.0	3.6	3.6	2.2	0.8	0.7	0.9		
(15) Precipitation	PrecAvg	42.6	0.4	0.0	0.9	1.5	7.0	3.9	3.6	5.1	9.3	8.5	1.9	0.3		
	PrecMax	54.1	3.7	0.2	2.4	3.7	12.2	7.2	8.9	12.4	17.2	18.7	7.7	2.5		
	PrecMin	30.1	0.0	0.0	0.0	0.0	1.8	1.1	1.7	1.8	3.6	2.4	0.0	0.0		
	PrecStd	6.7	0.9	0.1	0.9	1.1	2.9	1.9	1.9	2.7	4.0	4.4	2.1	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.4	100.7	104.2	106.7	104.9	98.3	95.0	95.0	95.0	94.1	95.2	93.5		
		MCWB	71.0	70.9	73.4	76.2	77.4	78.5	78.6	78.4	79.4	79.2	76.2	74.3		
	2%	DB	93.2	98.5	102.0	104.2	101.1	95.0	92.7	92.3	93.0	92.1	92.5	91.2		
		MCWB	71.3	71.4	73.0	76.0	78.1	78.8	78.3	78.2	79.0	78.8	76.4	73.9		
	5%	DB	91.2	96.5	100.3	102.3	98.2	92.7	90.8	90.4	91.2	90.6	90.3	89.3		
		MCWB	71.3	71.6	72.7	76.3	78.2	78.5	77.9	78.0	78.7	78.4	76.0	73.2		
	10%	DB	88.8	94.2	98.3	100.0	95.2	90.6	88.9	88.5	89.3	89.0	88.3	86.7		
		MCWB	70.9	71.7	72.7	76.3	78.0	78.3	77.5	77.6	78.4	78.0	75.4	72.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.1	78.3	80.1	81.2	81.8	81.7	80.9	80.9	81.8	81.4	80.1	78.4		
		MCDB	86.6	92.9	95.4	99.1	93.8	90.2	89.6	89.0	90.4	89.2	89.6	88.7		
	2%	WB	75.4	76.7	77.9	79.8	80.7	80.6	79.8	79.9	80.7	80.2	78.9	76.5		
		MCDB	85.5	90.7	92.8	96.5	92.5	89.7	88.6	88.1	89.3	88.6	87.7	86.3		
	5%	WB	74.1	75.4	76.7	78.9	79.8	79.8	79.0	79.1	79.8	79.6	77.9	75.3		
		MCDB	84.6	88.8	91.9	95.2	91.4	88.6	87.5	87.1	88.0	87.6	86.1	84.6		
	10%	WB	72.8	74.0	75.6	78.1	79.0	79.1	78.2	78.3	79.0	78.9	76.9	74.1		
		MCDB	83.8	87.7	90.8	93.7	90.3	87.6	86.3	86.0	86.8	86.6	84.5	82.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.3	25.5	21.6	19.6	14.3	11.5	10.4	10.6	12.2	13.7	17.4	21.2		
		MCDBR	27.5	27.8	23.9	21.4	18.3	14.3	12.8	13.2	14.5	16.1	20.2	23.4		
		MCWBR	9.4	8.5	7.2	6.5	5.2	4.9	4.8	4.8	5.3	5.5	6.7	8.1		
	5% WB	MCDBR	21.0	23.6	20.9	19.4	15.3	12.5	11.6	11.9	13.6	14.5	16.5	19.3		
		MCWBR	8.0	9.0	7.5	6.5	5.5	5.1	5.2	5.1	5.5	5.5	6.0	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.465	0.555	0.672	0.619	0.492	0.436	0.426	0.432	0.463	0.538	0.467	0.463		
	taud		2.134	1.910	1.659	1.829	2.222	2.403	2.428	2.400	2.295	2.035	2.212	2.160		
	Ebn at Noon		259	241	218	231	259	271	274	275	267	243	257	256		
	Edh at Noon		47	62	81	69	46	38	37	38	43	54	43	45		
(44) All-Sky Solar Radiation	RadAvg		1483	1669	1749	1890	1699	1475	1372	1359	1452	1464	1499	1426		
	RadStd		69	68	110	107	146	132	117	91	83	86	88	61		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	+0.29	N/A	N/A	N/A	N/A

Nomenclature: See separate page